

Forkable sandboxes let computers explore executable alternatives.

From scientific experiments to self-driving computers.

➤ Reuse reached state. Run alternatives. Verify the result.

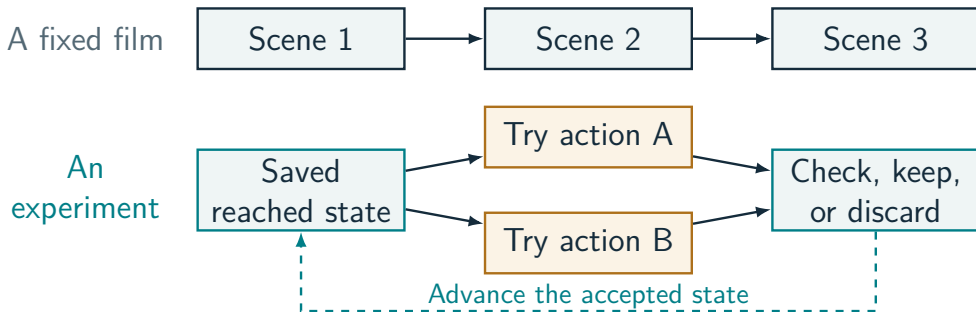
Yossi Eliaz

Principal Engineer, Incredibuild

Associate Professor, HIT

Cambridge Systems Research Group, 15 October 2026. Research and engineering perspectives; employer affiliation disclosed.

Forkable computation lets us try different futures from the same state.



Organizing analogy and proposed execution model. Stored-program hardware remains the substrate.

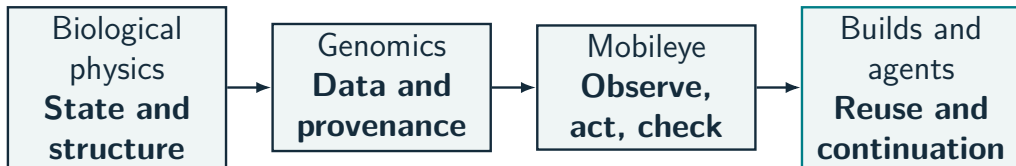
An adaptive controller chooses programs and continuations; IF and LOOP still execute them.

	Stored-program control	Self-driving execution layer
IF	Choose an encoded path	Evaluate alternative executions, then choose
LOOP	Repeat an encoded body	Revise the next action from observations
CALL	Execute a routine; return a value	Run a private continuation; return artifact and evidence

Checkpoint → fork → execute → evaluate → select → continue.

Proposed orchestration layer on existing stored-program machines.

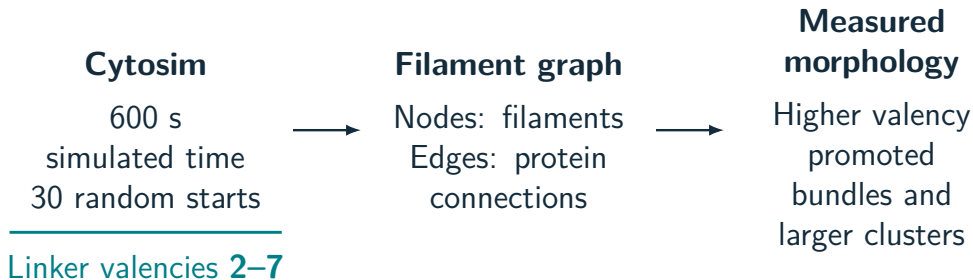
My work connects biological experiments, genomic pipelines, and autonomous execution.



What must remain valid when computation continues?

Physics PhD, 2020; coauthored genomics work; Mobileye perception, 2021–2024; current Incredibuild and HIT roles.

Higher linker valency changed simulated actomyosin structure.



The binding configuration changed the collective structure.

Eliaz, Nedelec, Morrison, Levine & Cheung, [Phys. Rev. E 102:062420 \(2020\)](#). First-author paper with coauthors.

A sparse algorithm made 500-bp chromosome analysis feasible.

Chromosome 1 A/B compartment calculation at 500-bp resolution

POSSUMM

Sparse matrix-vector operations

- **23 GB RAM**

2.5 min

Measured runtime

Dense approach

Projected memory requirement

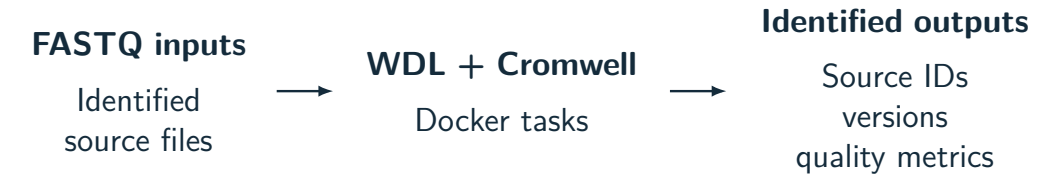
> 4.6 TB 



Harris et al., including Eliaz, [Nature Communications 14:3303 \(2023\)](#). Team result.

ENCODE linked reproducible execution to identifiable outputs.

14,000+ datasets

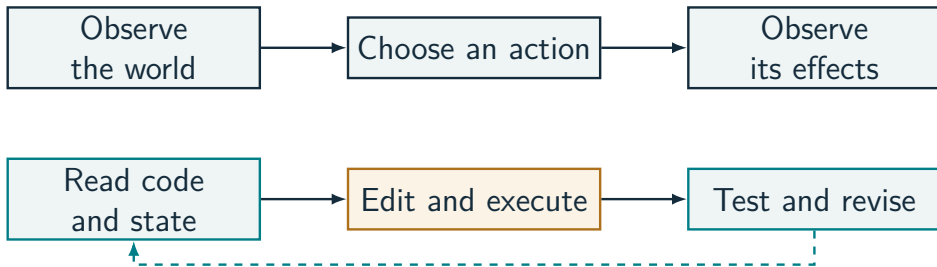


Each output records how the analysis produced it.

Hitz et al., including Eliaz, [The ENCODE uniform analysis pipelines \(2023\)](#). Coauthored infrastructure.

A self-driving computer closes a feedback loop around its own actions.

Mobileye: perception work, 2021–2024



Personal work: Mobileye perception. Lower lane is the software-agent analogy.

Reusing build artifacts reduced a reported Tokio compile from 46 to 13 seconds.

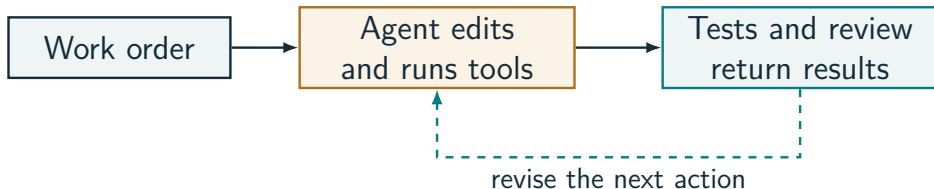
Fresh compile  ~46 s

Restored cache  ~13 s

Hot cache  ~3 s

Eliaz, [Tokio issue #8200](#), June 2026: external Incredibuild demo. Artifact caching.

An agent changes the environment that executes its next experiment.



Our Airflow Factory uses Docker sandboxes and hosted models; the recorded work orders ran serially.

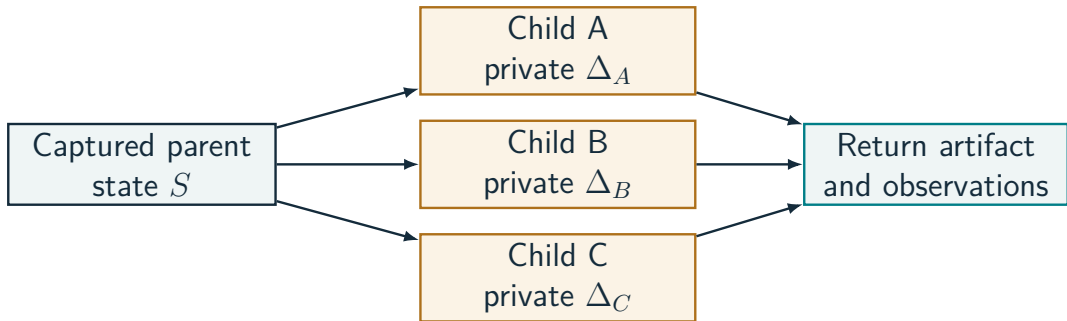
Alternative experiments can share the work already completed.



Can we branch from the state
we already paid to reach?

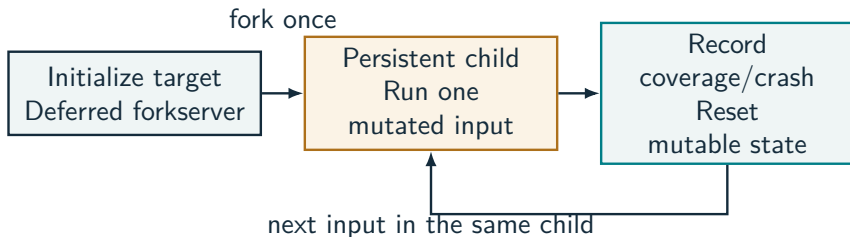
Design question: reusable reached state, independent private changes.

A fork creates private continuations from one captured execution state.



Execution abstraction; backend-specific state and external-effect semantics.

AFL++ forks initialized state once and resets between inputs.



About 1,000 inputs per child, then restart the child.

[AFL++ persistent-mode documentation.](#)

AFL++ reports 10–20 times faster execution in persistent mode.

10–20×

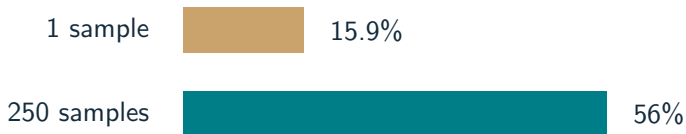
Typical speedup reported for persistent mode

One child amortizes process creation across many inputs.

Correct per-input reset keeps the reused state useful.

[AFL++ documentation](#); published typical gain, not our benchmark.

250 samples raised SWE-bench Lite coverage from 15.9% to 56%.

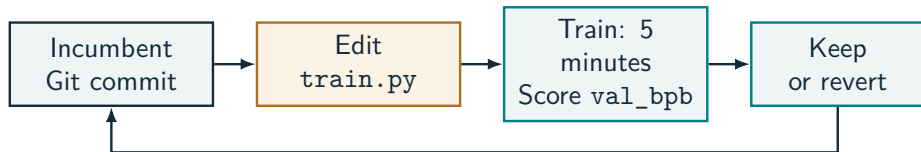


DeepSeek-Coder-V2-Instruct; pass at least once

Coverage means a correct patch exists; a selector must recognize it.

Brown et al., Large Language Monkeys (2024); SWE-bench Lite.

Autoresearch kept 23 experiments and lowered validation bits per byte.

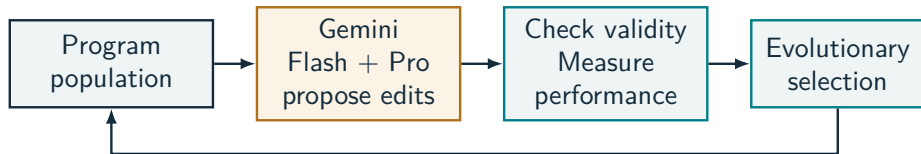


126 attempts: 23 kept, 102 discarded, 1 crash.

0.997900 $\rightarrow$ 0.969686: a 2.8% reduction; lower is better.

Karpathy, autoresearch protocol; reported session.

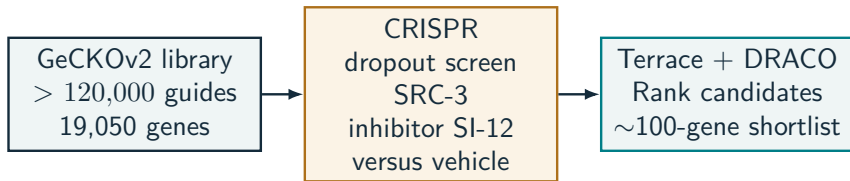
AlphaEvolve recovered 0.7% of Google's worldwide compute.



A deployed Borg scheduling heuristic recovered 0.7% on average.

Google DeepMind, AlphaEvolve (2025); production application.

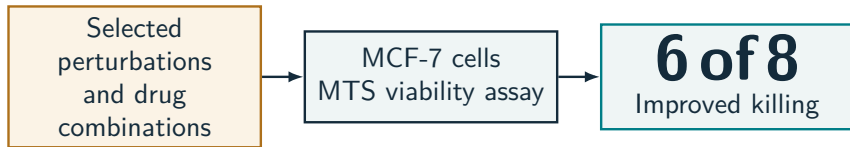
A genome-scale screen ranked candidates for sensitivity to SI-12.



We sought perturbations that increase cancer-cell sensitivity to SI-12.

Gilad, Eliaz et al., *Communications Biology* 4:399 (2021); screen and candidate ranking.

Six of eight combinations improved killing in MCF-7 cells.



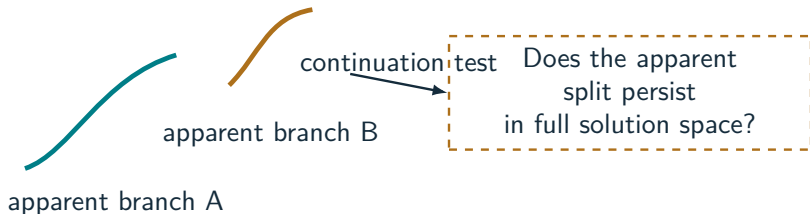
Each point: ≥ 4 technical replicates. Each plot: ≥ 2 independent experiments.

Gilad, Eliaz et al. (2021), Figs. 2–4; in-vitro MCF-7 assays.

We tested whether projected three-body orbit branches connect.

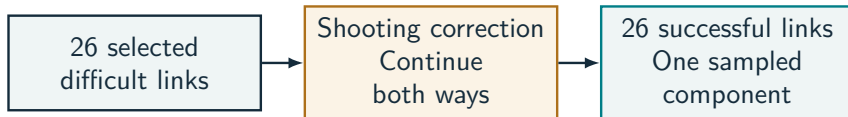
With Ori Chamo: 135,445 catalogued unequal-mass periodic three-body orbits.

Projected view (schematic)



Chamo–Eliaz, [ai.viXra:2608.0069](#) (2026); preliminary three-body continuation study.

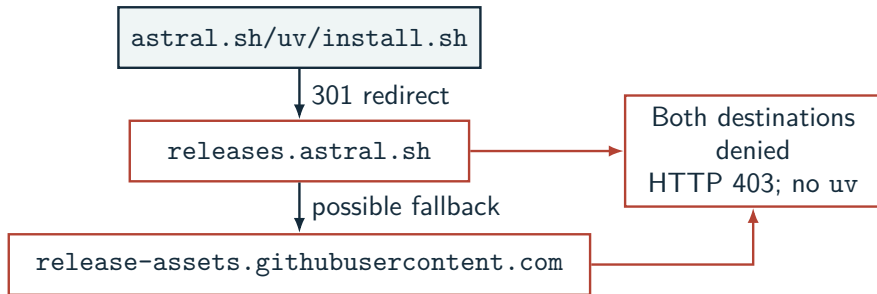
All 26 selected links connected by bidirectional continuation.



Independent 60-digit checks confirmed representative Floquet stability transitions.

Chamo-Eliaz preprint; Three-Body Orbit Atlas records.

The installer fix added two missing hosts across a six-file plan.



Airflow + Docker Sandboxes; six files across policy, scripts, tests and docs.

SELFHOST-2 intent, plan and records; 27 September 2026.

The factory run took 43:44 and cost \$10.33 in model calls.

Recorded stage	Time	Model cost
Specification and plan	3:41	\$0.96
Build and test	20:01	\$4.50
Review	18:55	\$4.87

43:44 elapsed; 42:38 in the recorded stage totals.

SELFHOST-2 metrics; agent-session durations and reported list-price model cost.

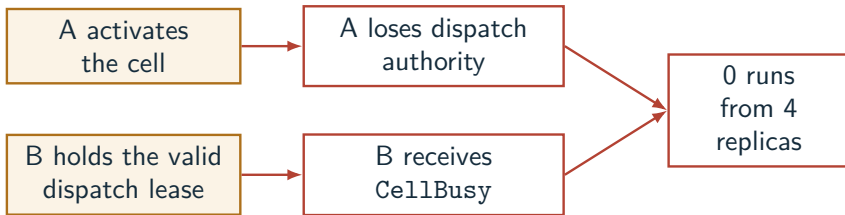
Two repairs and two reviews used 59% of time and 75% of cost.



Both bars refer to two repairs plus two reviews

SELFHOST-2 metrics; repair and review session records.

The repair agent diagnosed a split between cell and dispatch ownership.



Repair: adopt the active cell when it belongs to the same work order.

SELFHOST-2 first repair diagnosis; dispatch-concurrency test.

The green suite left same-owner adoption without a direct test.

Required deterministic check

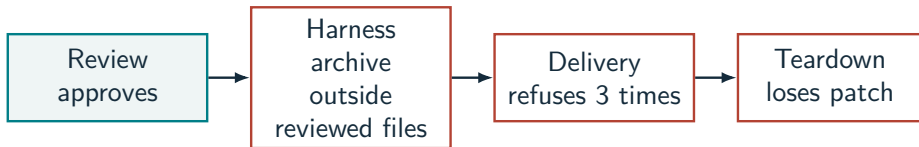


Suite: 1,972 reported cases, including skips; zero failures.

Review requested this test; the direct deterministic check stayed absent.

SELFHOST-2 final review; repairs and harness test records.

Three delivery refusals preceded cleanup that erased an approved patch.

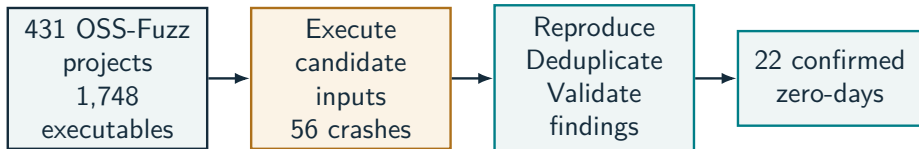


Separate work order: about 26 minutes, \$3.43, no delivered PR.

DESIGN Export the candidate and receipts before teardown.

SELFHOST-3 delivery records; recovery design. Separate from the SELFHOST-2 installer task.

CyberGym validated 22 zero-days after a 56-crash campaign.



OpenHands + GPT-5 proposed inputs; sanitizer runs produced candidate crashes.

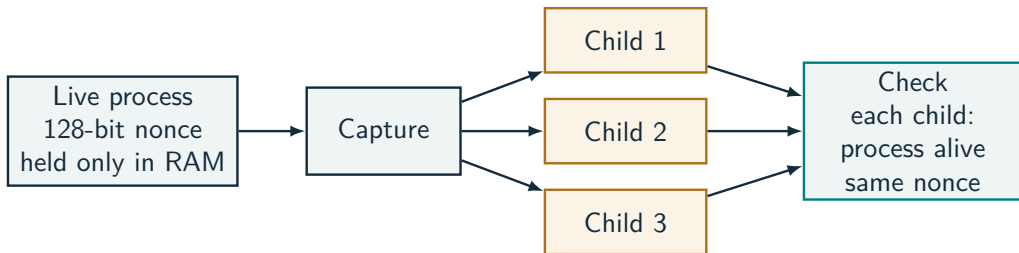
[Berkeley RDI](#), [CyberGym](#); [CyberGym report](#); latest-code campaign.

The next action determines which state a continuation needs.

Next action	Mechanism	State received
Edit source or build	Worktrees and build cache	Files and cached inputs
Continue live execution	Process or VM checkpoint	Supported memory and runtime state
Reuse an observation	Replay	Recorded execution or response

CRIU, gVisor, Firecracker and rr documentation; SnowFlock, EuroSys 2009; DeepSeek-V4 §5.2.5.

A RAM-only nonce tests memory restoration in three children.



Experiment design: preregistered Gate 0; Kimi K3 §5.3.2; CRIU and Firecracker snapshot documentation.

A checkpoint and a remote fork time different operations.

System	Reported operation	Latency
DeltaBox	Checkpoint	10.83 ms
SnowFlock	Remote fork across hosts	600–800 ms

DeltaBox; Lagar-Cavilla et al., SnowFlock, EuroSys 2009. Full operation table retained in QA.

islo completed 255/256 restore–run–capture calls at a 6.87 s median.

Client-observed trace	Recorded value	
Operation	Restore, run command, capture	
Completed calls	255 / 256	
Latency	p50 6.87 s	p95 9.04 s
Input and concurrency	Named 141 MB snapshot; 12 concurrent calls	

Eliaz, [arXiv:2607.09689v4](#), Table 2. Teardown excluded from percentiles.

Reuse saves resources when avoided preparation exceeds its overhead.

$$(N - 1)P > H + N(R + D)$$

P : preparation per continuation H : capture

R : restore D : incremental divergence N : children

Additive resource cost; common continuation work cancels.

Illustrative resource-cost model; Jitsu, NSDI 2015; LightVM, SOSP 2017.

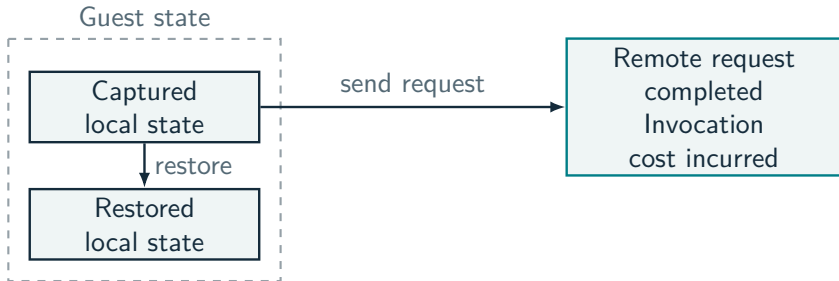
A warm cache can reverse the resource advantage of restoration.

Illustrative calculation: $N = 8$, $H = 2$, $R = 1$, $D = 3$ resource-seconds.

Preparation	Avoided	Reuse overhead	Net saving
$P = 30$: cold preparation	210	34	+176
$P = 2$: warm cache	14	34	-20

Illustrative parameter values; the same capture, restore and divergence costs in both rows.

Restoring a guest cannot undo a completed remote model request.



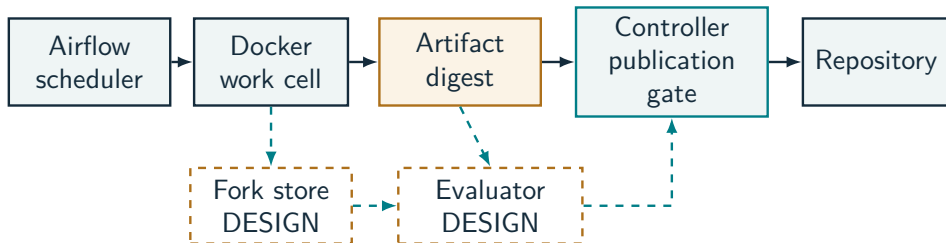
Speculator, SOSP 2005; external synchrony, OSDI 2006; Remus, NSDI 2008; DeepSeek-V4 §5.2.5.

The controller holds the credential that publishes a child's candidate.



Factory artifact-digest gate; Firecracker and gVisor documentation; Capsicum, USENIX Security 2010; CHERI, IEEE S&P 2015.

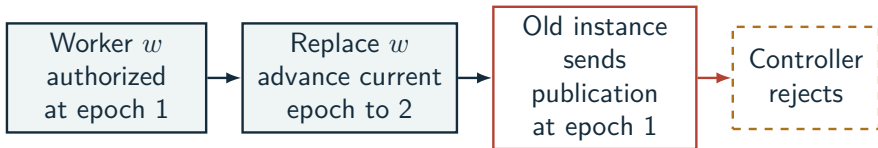
The factory routes candidate digests through a controller publication gate.



Solid: exercised serial components. Dashed: extension design.

Factory records and design; Eliaz, arXiv:2607.09689v4.

The controller rejects an old epoch after replacing a worker.



Epoch-fenced publication design; Gray & Cheriton, Leases, SOSP 1989.

Selecting, composing, and pooling results require different checks.

Returned objects	Decision	Required check
Different patches	Select one	Validate that artifact
Compatible patches	Compose	Test the exact composition
Measurements of one quantity	Pool	Justify information and dependence

Factory records; numerical reducer, arXiv:2607.09689v4.

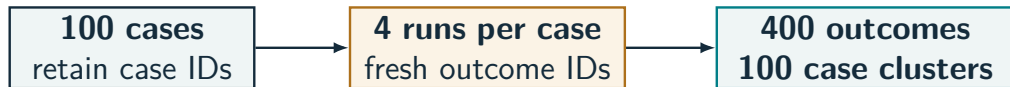
Passing every pair does not establish that all three patches pass.

Each patch enables one worker. Capacity permits two.

Applied patches	Workers	Capacity check
Each single patch	1	✓ pass
Every pair	2	✓ pass
All three	3	✗ fail

Exact illustrative capacity example; test the selected composition.

Four runs of 100 cases produce 400 outcomes in 100 case clusters.



A fresh outcome retains its case and ancestry.

Illustrative evidence ledger; analogous replicate distinction in the genomic assay.

Linking Bitcoin addresses recovered 64 agents behind most early mining.

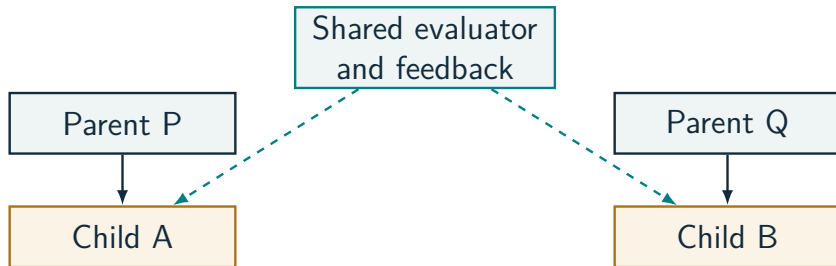


3 January 2009–9 February 2011

Distinct identifiers can conceal a common source.

Blackburn, Huber, Eliaz et al., [arXiv:2206.02871](https://arxiv.org/abs/2206.02871) (2022), coauthored study.

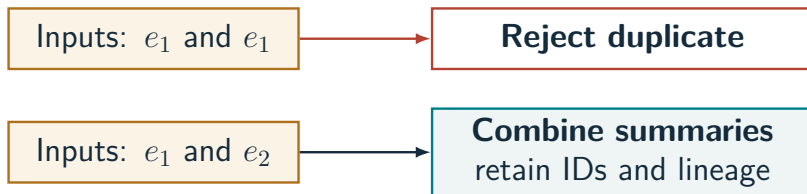
Different parents can still share the evidence that guides their children.



Solid: execution ancestry. Dashed: shared information.

Proposed execution manifest: ancestry, fixtures, evaluator and feedback links.

The reducer rejects a repeated evidence ID and retains distinct evidence.

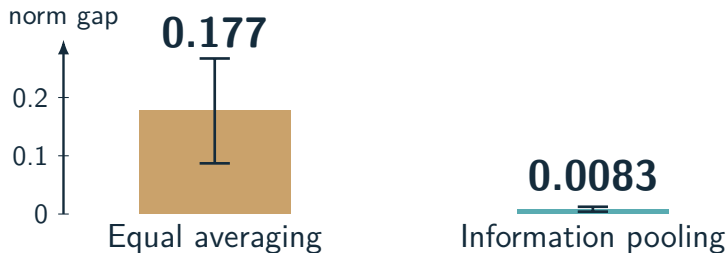


Model: common parameter, calibrated information, independent noise.

Built reference reducer; arXiv:2607.09689v4.

Information pooling reduced the synthetic MLE gap from 0.177 to 0.0083.

Five logistic-regression shards: $n = 60, 120, 400, 2,000, 5,000$.

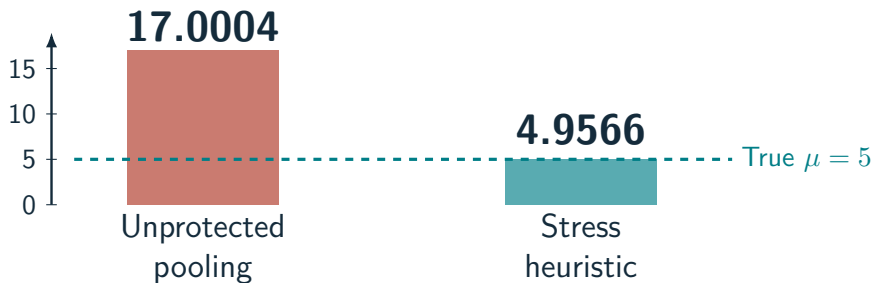


Eight fixed seeds; whiskers are SD. Lower distance to the full-data MLE is better.

Synthetic check, Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689) (2026), common-parameter model.

A forged precision report moved the synthetic pooled estimate from 5 to 17.

One 2,000-point shard targets 17 and inflates information per point 50 $\times$.



Eliaz, [primary demo.py](#) and arXiv:2607.09689. Synthetic stress case; no Byzantine guarantee.

At correlation 0.1, 100 measurements have the mean precision of about nine.

$$N_{\text{eff}} = \frac{N}{1 + (N - 1)\rho}$$
$$\frac{100}{1 + 99(0.1)} = 9.17$$

Equal marginal variance and common pairwise correlation.

Exact variance calculation for a mean; illustrative correlation value.

Zero pairwise correlation can coexist with an all-wrong rate of 25 percent.

Three candidates; 1 means wrong. Listed vectors are equally likely.

Error vectors	Each wrong	Pair correlation	All wrong
$\{000, 011, 101, 110\}$	50%	0	0%
All eight vectors	50%	0	12.5%
$\{001, 010, 100, 111\}$	50%	0	25%

A selector returning one candidate succeeds at most $1 - P(\text{all wrong})$.

Exact enumeration; selector ceiling also discussed by Chen, arXiv:2606.27288.

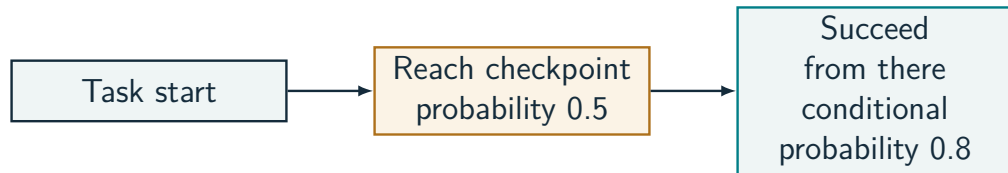
Shared variation can make a paired comparison more precise.

$$\begin{aligned} & \text{Var}(A - B) \\ &= \text{Var}(A) + \text{Var}(B) - 2 \text{Cov}(A, B) \end{aligned}$$

Compare A and B on the same case and controlled conditions.

Covariance identity; common-random-numbers methodology.

Success after reaching a checkpoint does not measure success from task start.



End-to-end success: $0.5 \times 0.8 = 0.4$

Illustrative conditional-probability example; Go-Explore motivates returning to reached states.

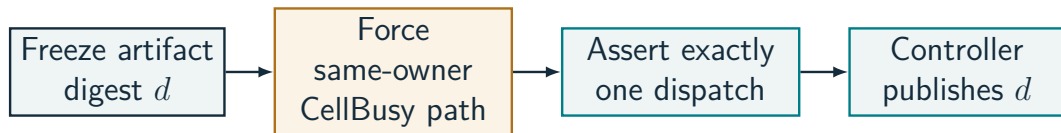
A receipt must describe the operation that actually executed.

Recorded label	What the record established
CI success	Evaluation skipped; job ended in 3 seconds
Human approval	Harness answered the approval gate
Adapter digest	Shim source outside Git; bytes unavailable

Record the actor, executed check, artifact digest, and durable bytes.

Our factory logs and repository audit: three distinct receipt failures.

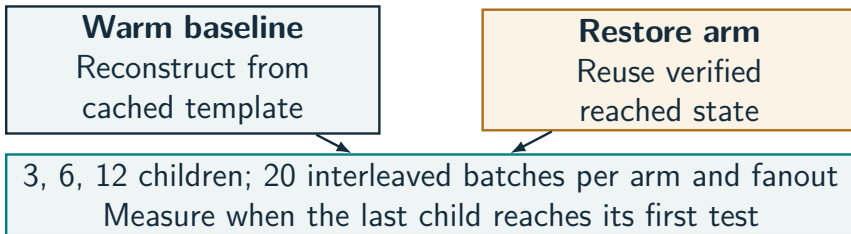
Publication must follow a passing check of the exact artifact.



The candidate cannot authorize its own publication.

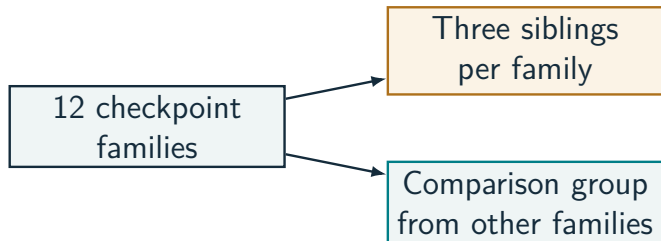
Acceptance design tied to the recorded adoption bug; reducer and gate are separate components.

The runtime experiment compares faithful restore with warm reconstruction.



Proposed H1 protocol, PREREGISTRATION.md; state-fidelity gate precedes timing.

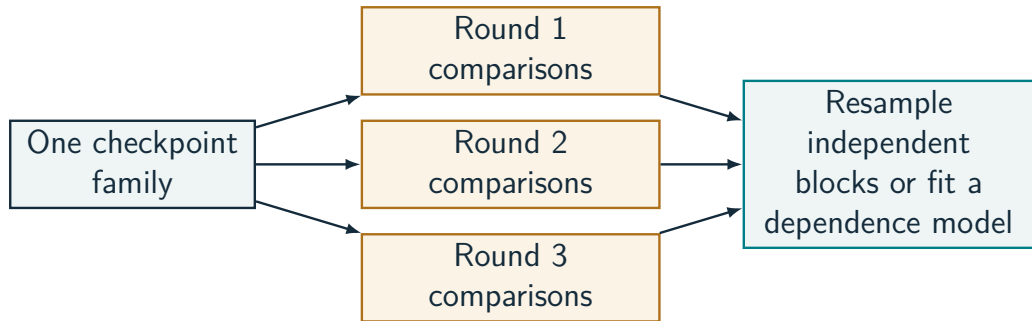
The cofailure experiment compares siblings with other checkpoint families.



At least 300 rounds
joint failures
and consumed state

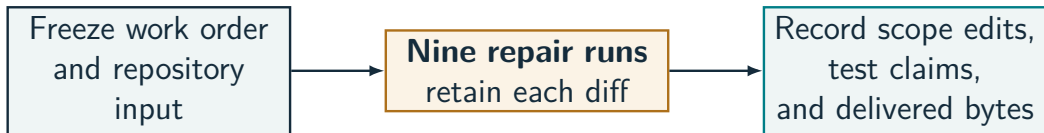
Proposed H2 protocol; causal treatment is a consumed parent-state feature.

Overlapping families and repeated rounds require grouped analysis.



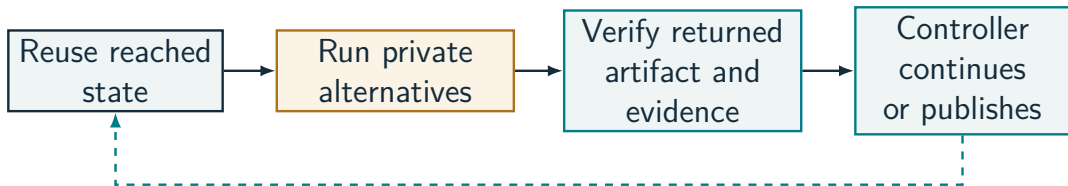
Analysis amendment required for overlap and repeated time measurements.

Nine fixed-input repairs test whether the recorded behavior repeats.



Proposed repeated-agent experiment; exact-artifact gate remains a separate deterministic check.

A useful continuation returns its artifact, evidence, and authority context.



Self-driving computers need reusable execution
and an explicit acceptance boundary.

Integrated research design connecting prior science, build reuse, factory incidents, and evidence-aware reduction.